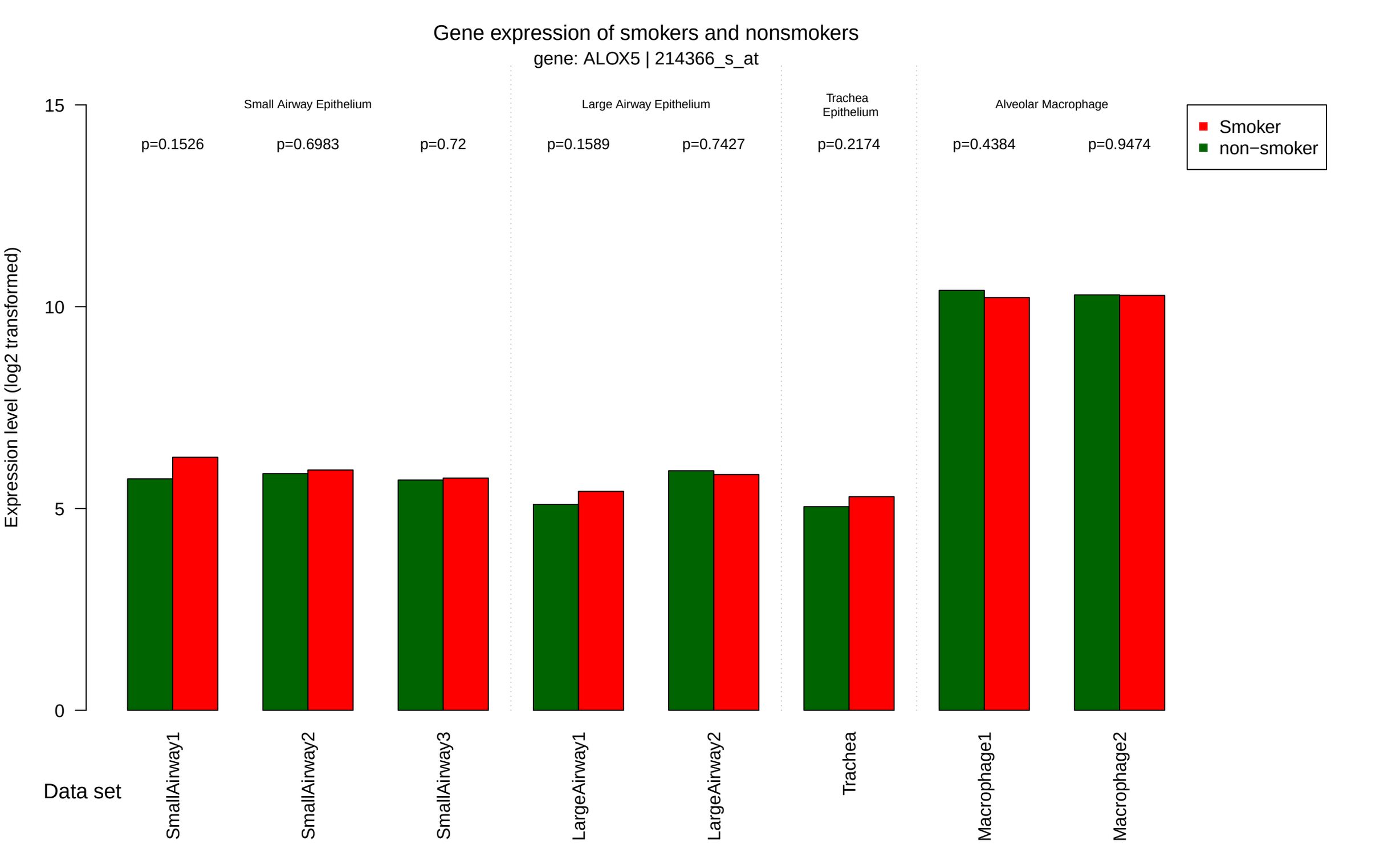
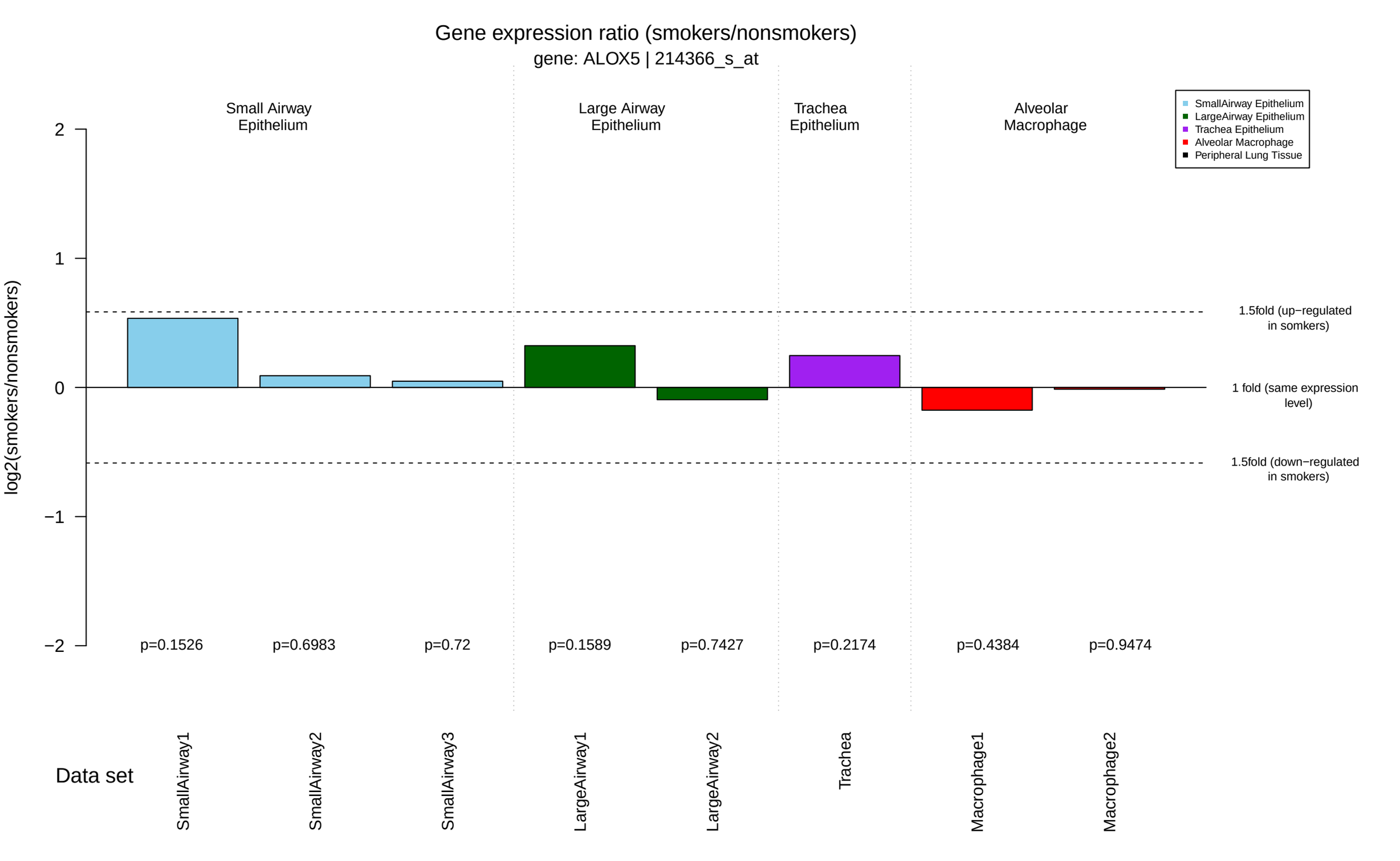
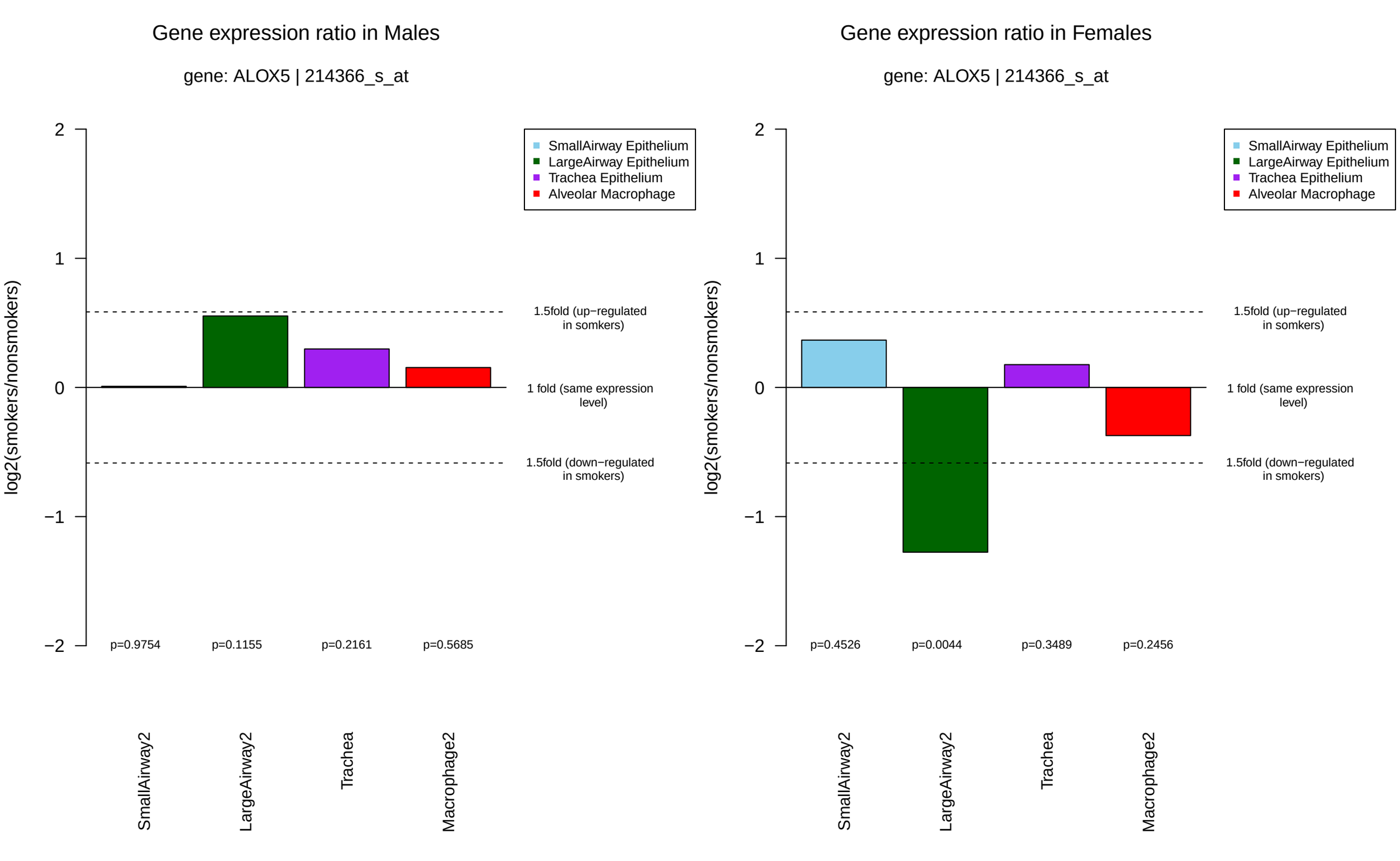
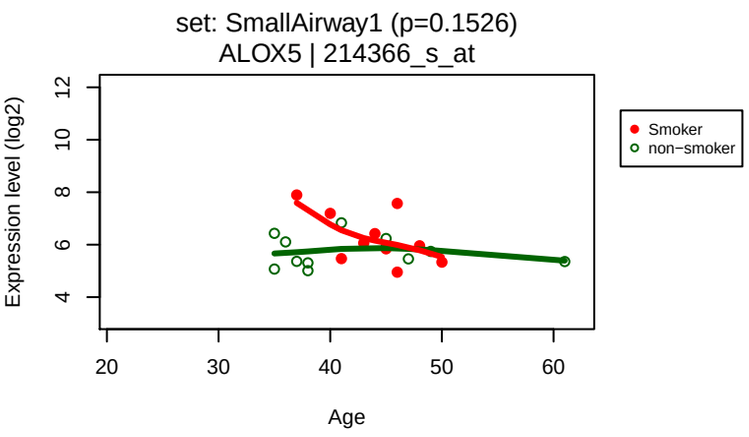
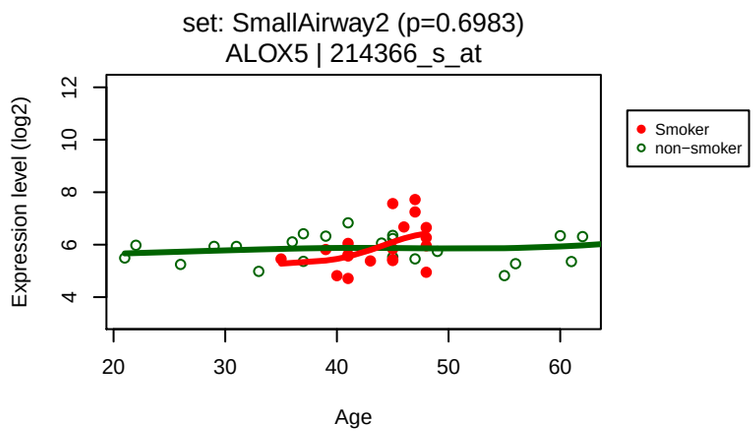
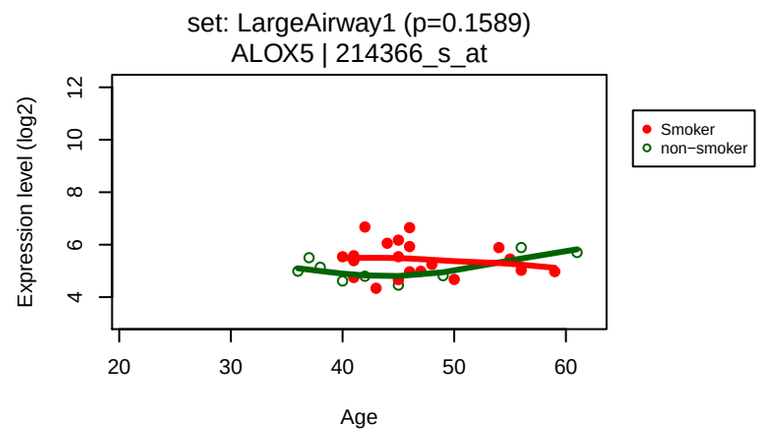

Gene expression of smokers and nonsmokers
gene: ALOX5 | 214366_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1526
p=0.6983
p=0.72
p=0.1589
p=0.7427
p=0.2174
p=0.4384
p=0.9474
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

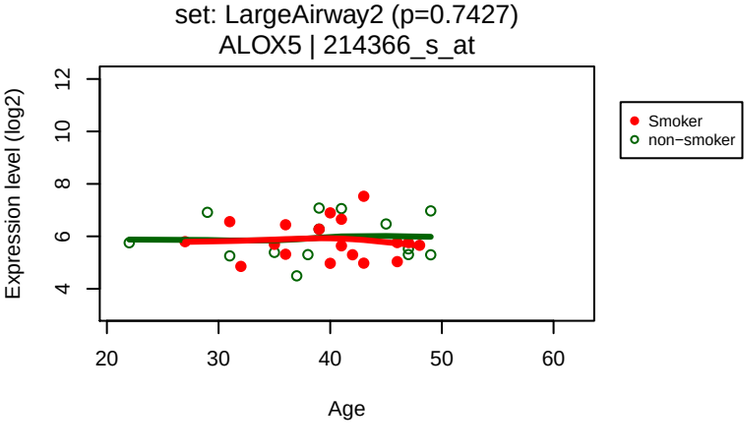

Gene expression ratio (smokers/nonsmokers)
gene: ALOX5 | 214366_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1526
p=0.6983
p=0.72
p=0.1589
p=0.7427
p=0.2174
p=0.4384
p=0.9474
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

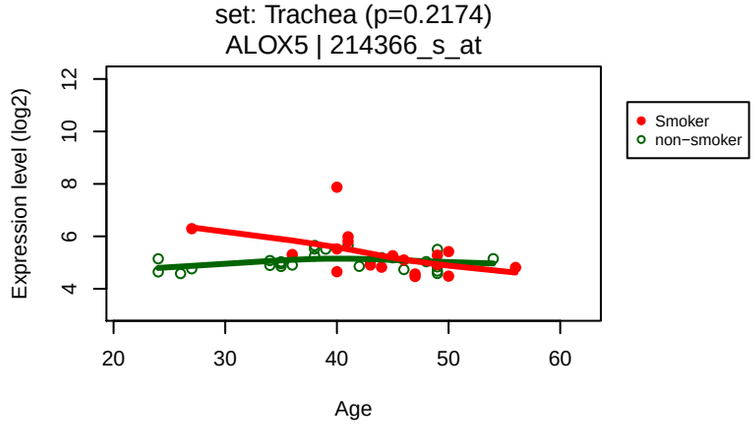

Gene expression ratio in Males
gene: ALOX5 | 214366_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9754
p=0.1155
p=0.2161
p=0.5685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX5 | 214366_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4526
p=0.0044
p=0.3489
p=0.2456
SmallAirway2
LargeAirway2
Trachea
Macrophage2

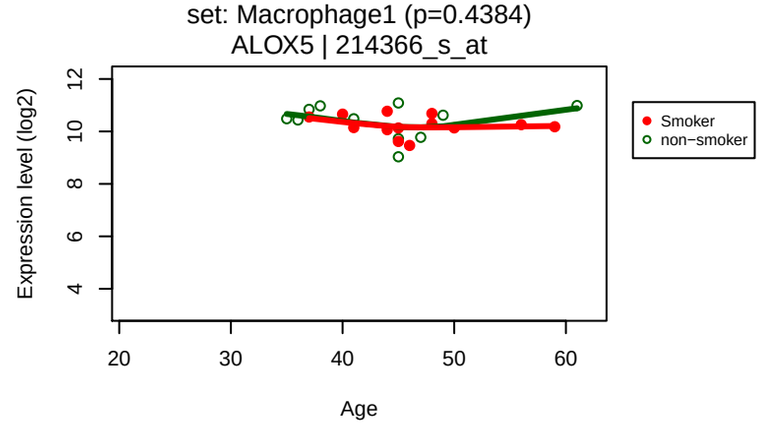

set: SmallAirway1 (p=0.1526)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker

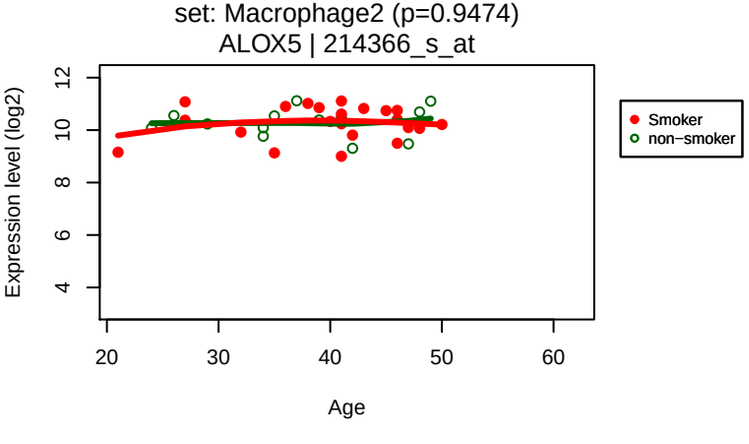

set: SmallAirway2 (p=0.6983)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker

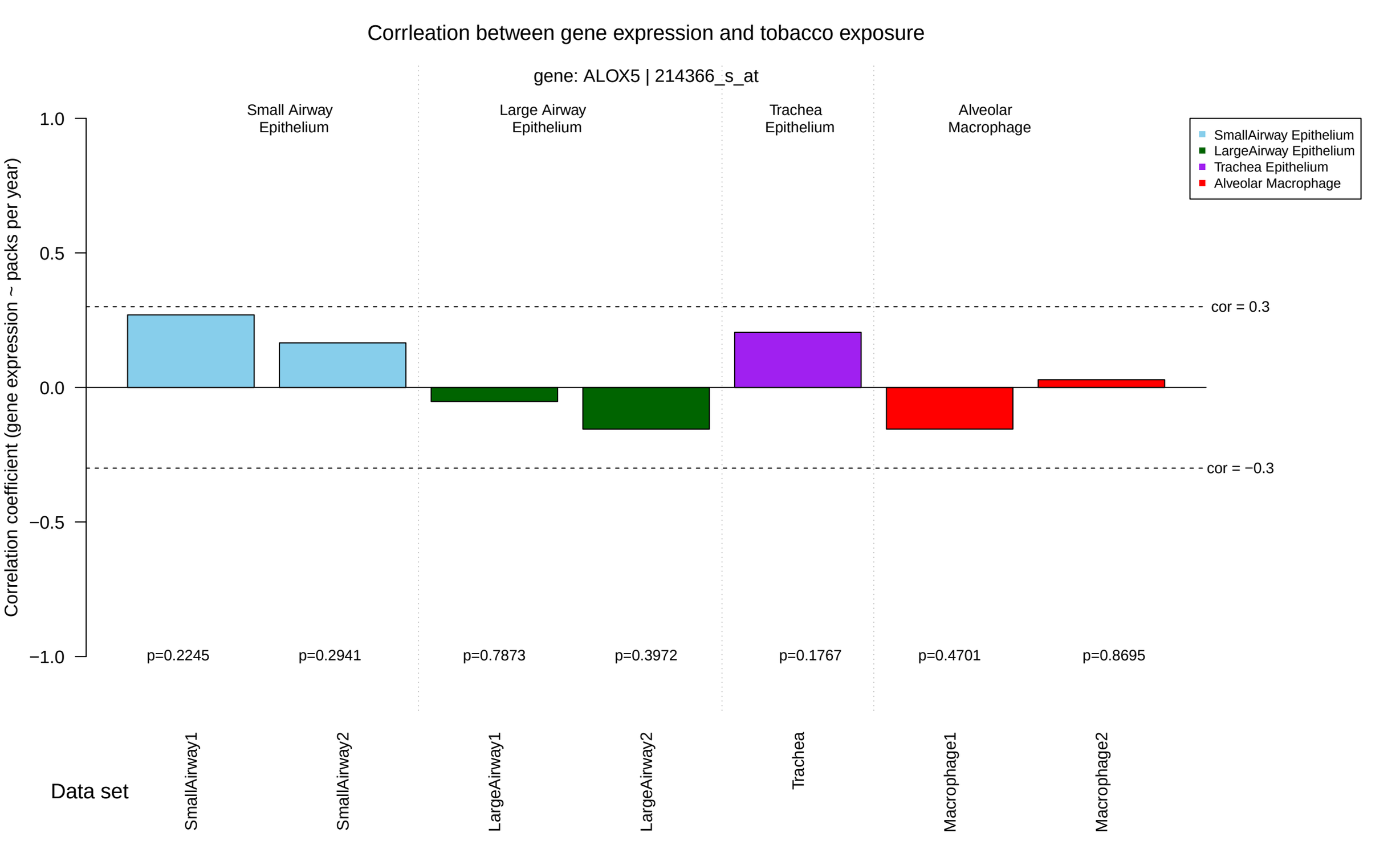

set: LargeAirway1 (p=0.1589)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7427)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2174)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4384)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9474)
ALOX5 | 214366_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX5 | 214366_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2245
p=0.2941
p=0.7873
p=0.3972
p=0.1767
p=0.4701
p=0.8695
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2